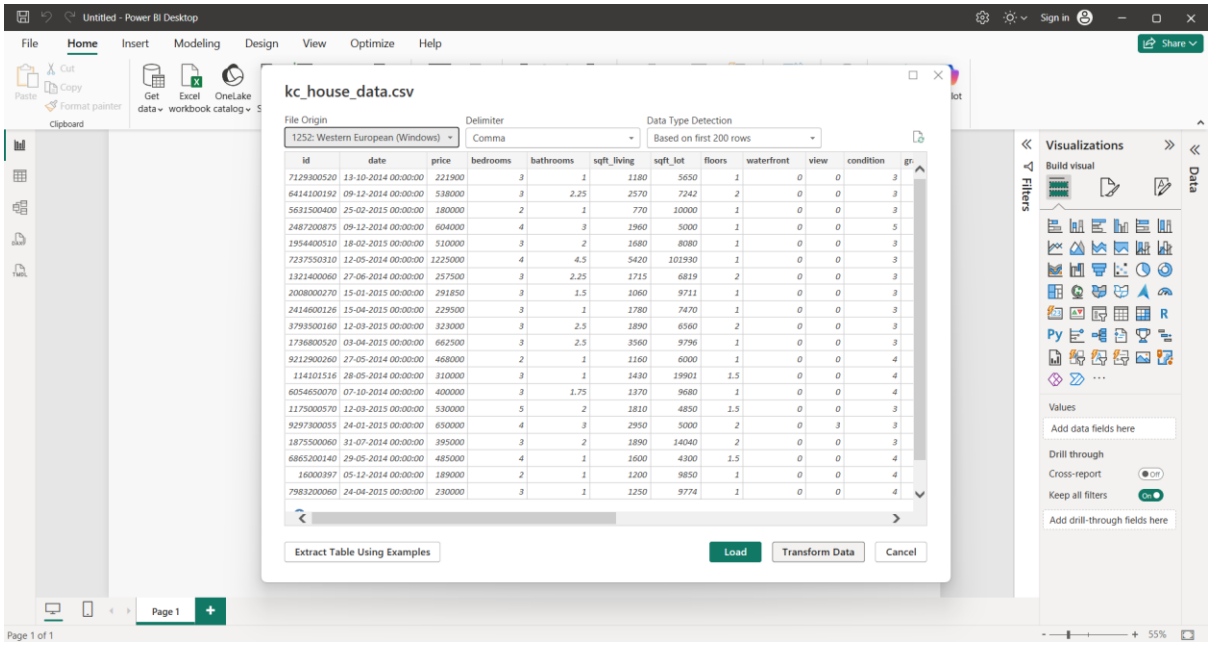
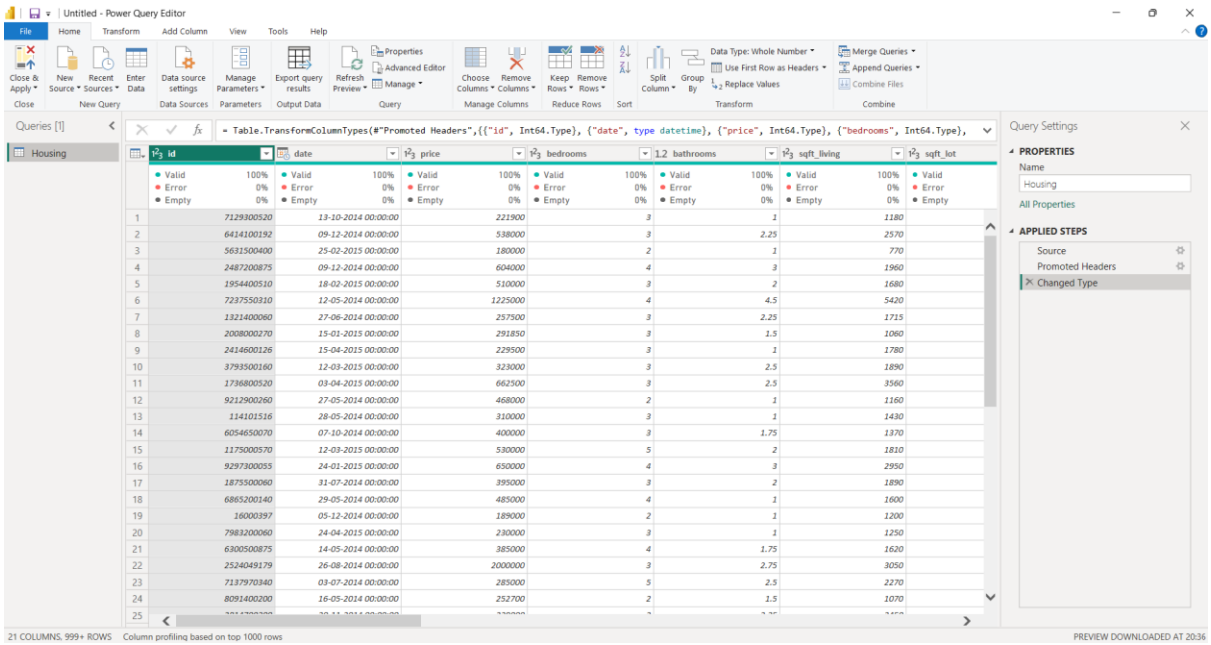
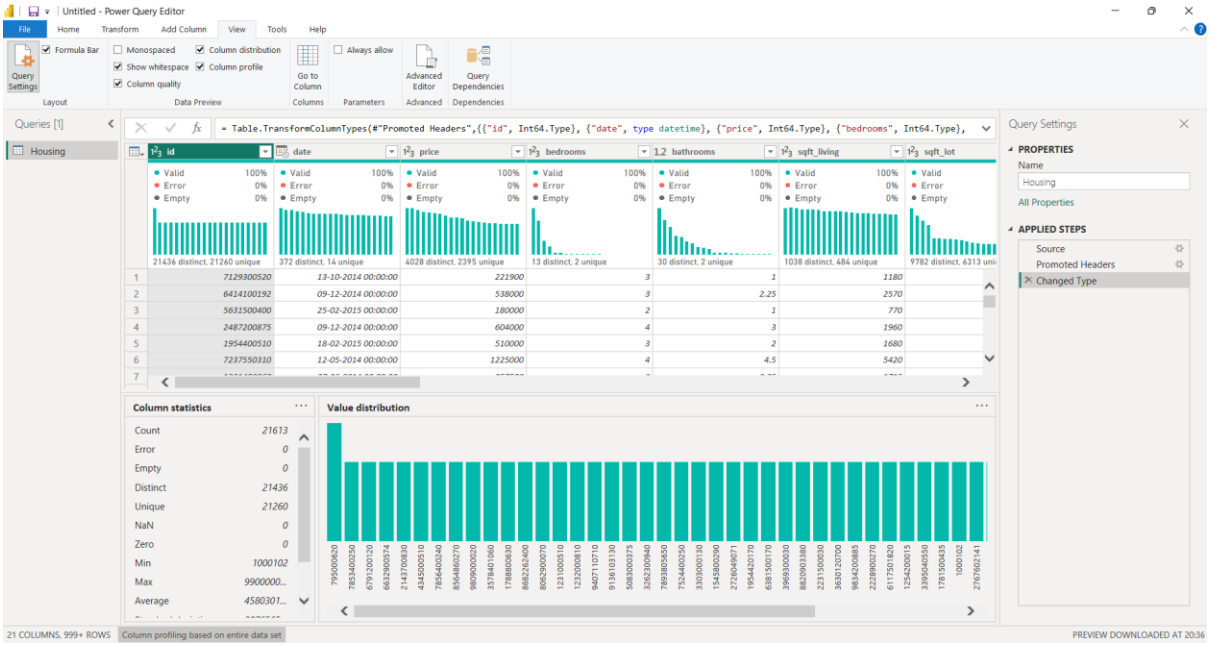
WEEK 9 SAI SHRUTI B



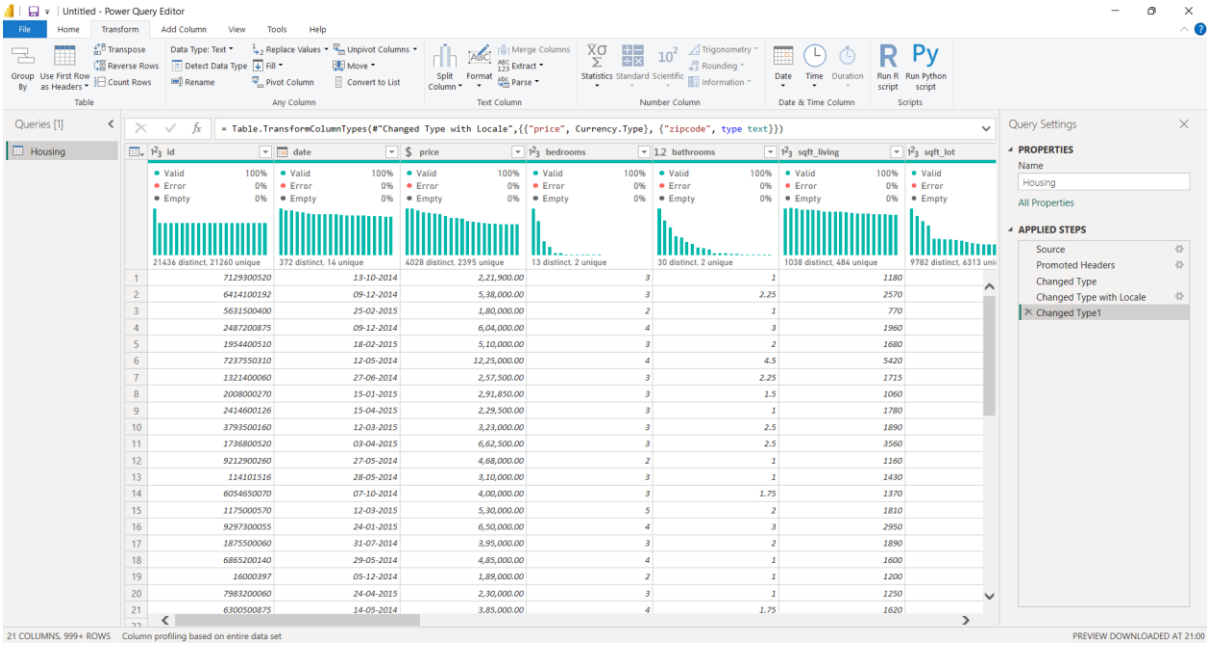
The kc_house_data.csv dataset was imported into Power BI. The data preview displays housing information, including property ID, sale date, price, bedrooms, bathrooms, living area, and lot area. The Transform Data option was selected to clean and prepare the dataset before loading it.



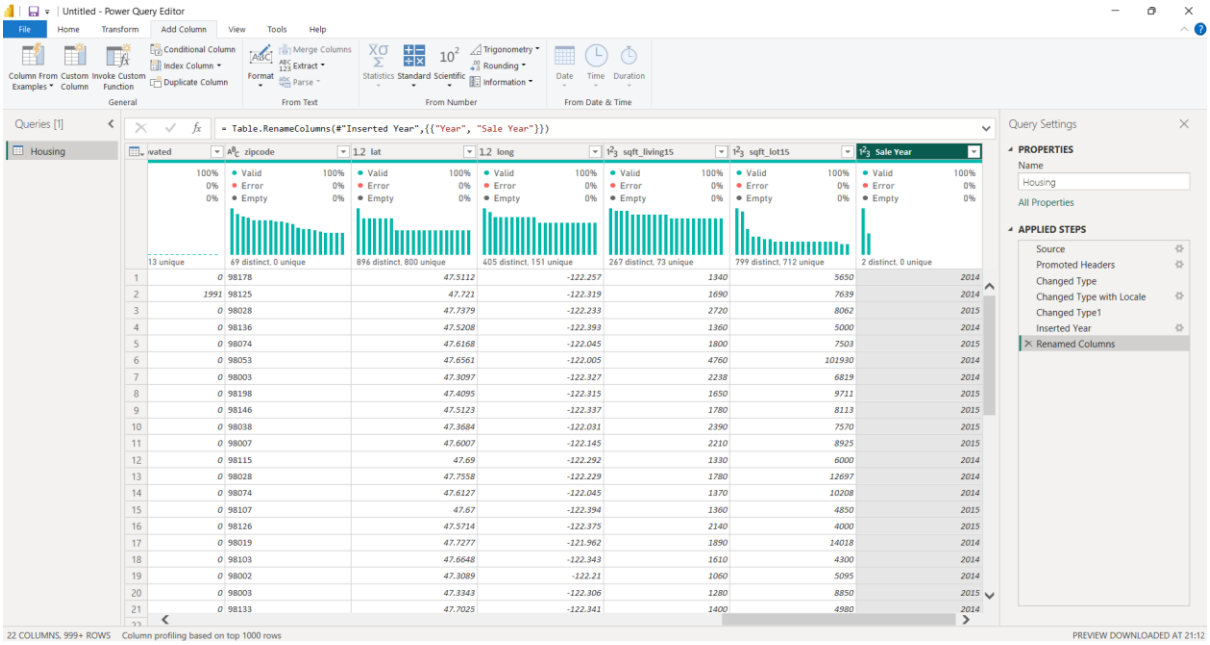
The housing dataset was opened in Power Query Editor. The column headers were promoted, and data types were assigned to the columns. The table was named Housing, and the initial data preparation steps were displayed in the Applied Steps pane.



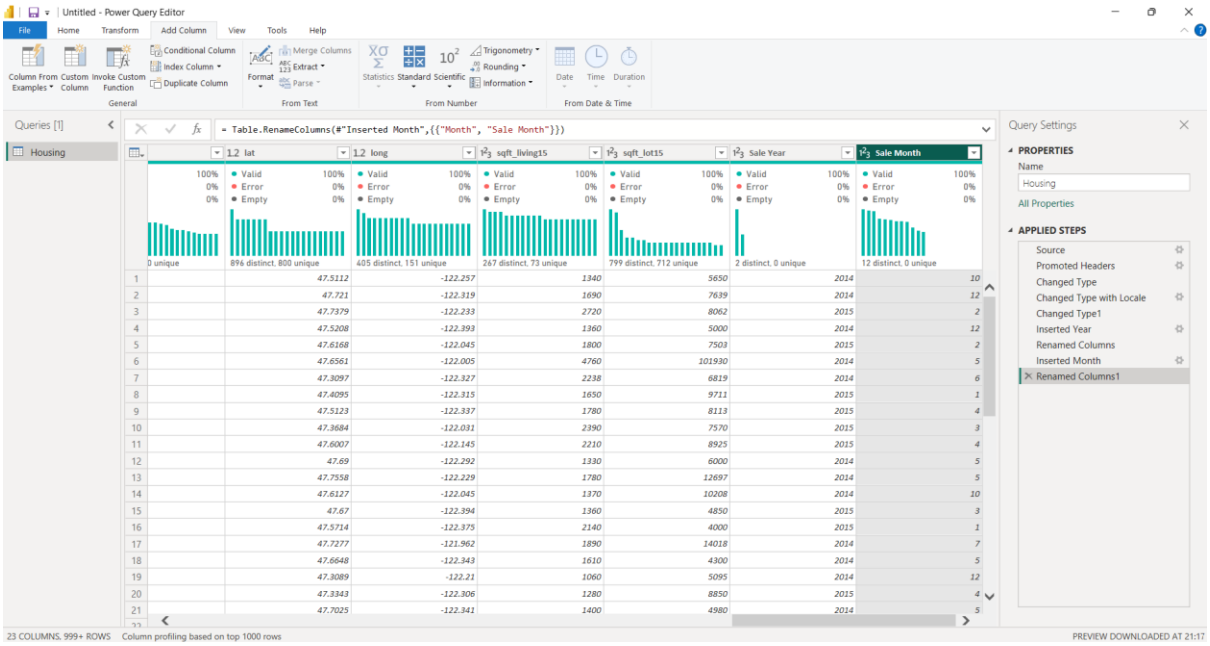
Column quality, column distribution, and column profiling were enabled to inspect the dataset. These features display valid, error, and empty values, along with distinct and unique counts. This helped assess the data quality before performing further transformations.



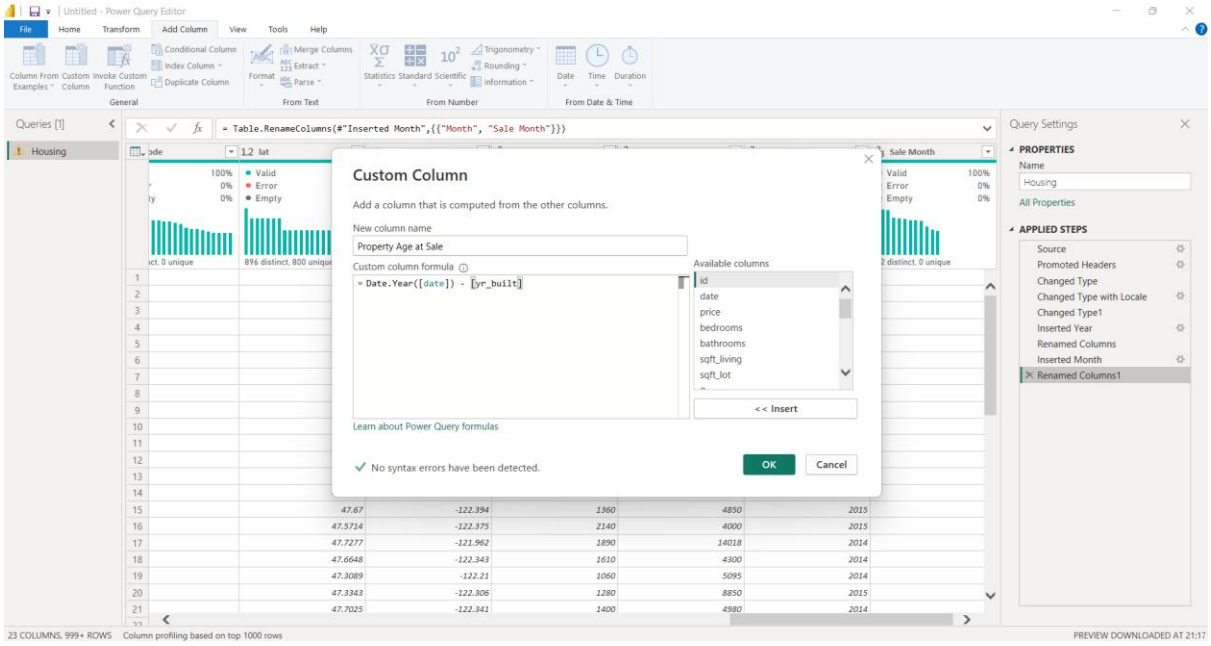
The data types of the housing columns were reviewed and adjusted. The date column was converted to a date/time type, the price column was formatted as currency, and the zipcode column was set to text to preserve postal-code formatting.



The year was extracted from the date column using Power Query's date transformation. The resulting column was renamed Sale Year, allowing the year of each property sale to be analysed separately.



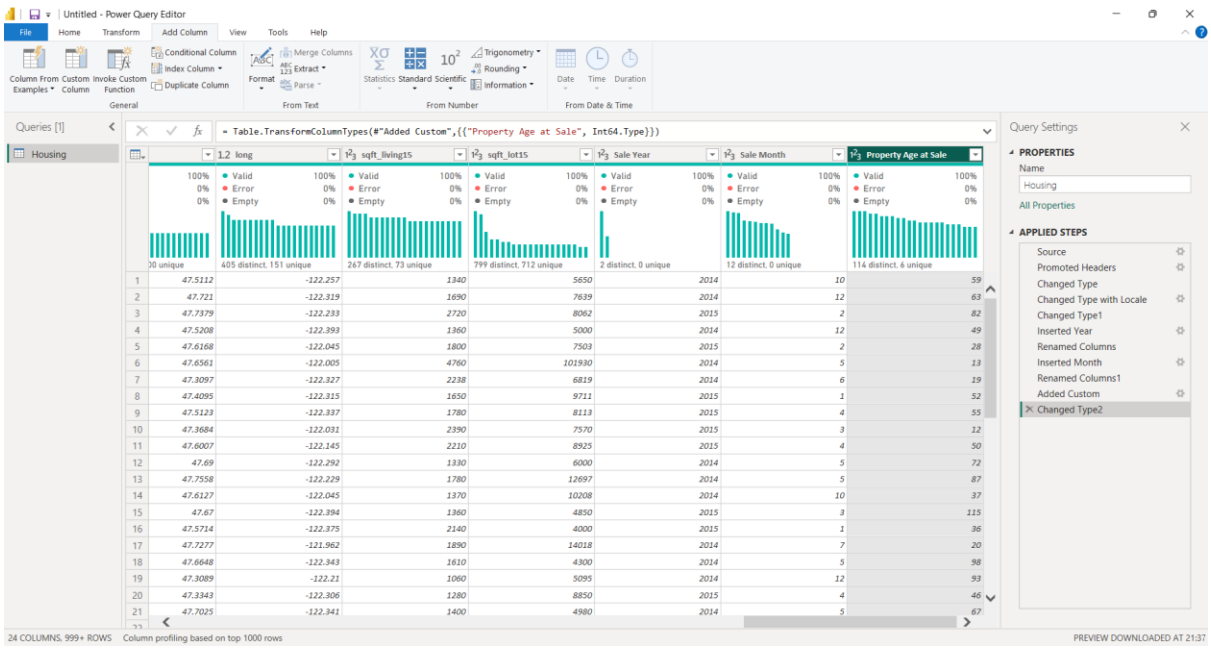
The month number was extracted from the date column using the date transformation tools. The resulting column was renamed Sale Month, allowing sales to be grouped and analysed by month.



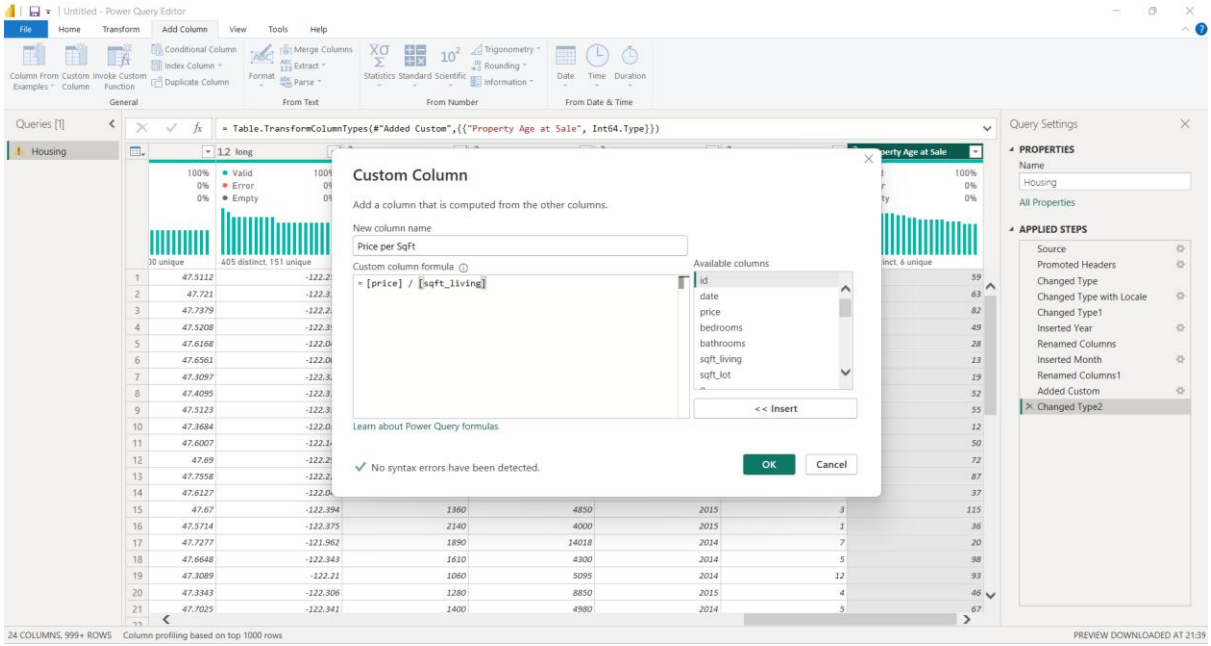
A custom column named Property Age at Sale was created using the formula:

Date.Year([date]) - [yr_built]

This calculates the property's age at the time of sale by subtracting the construction year from the sale year.



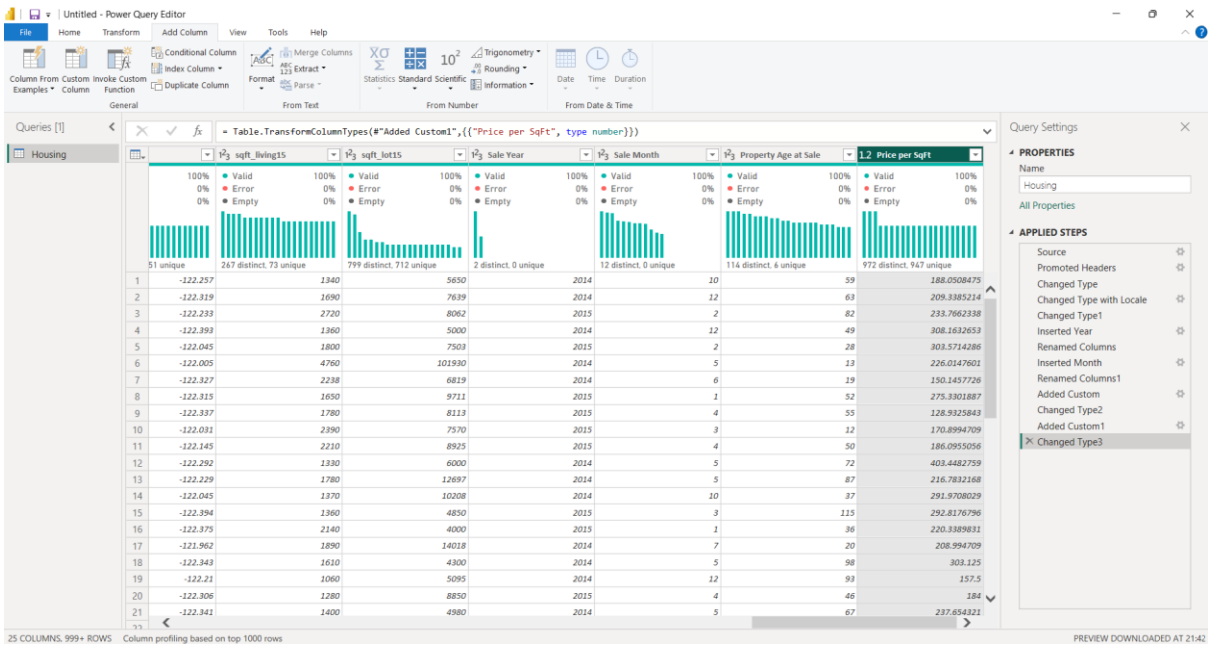
The newly created Property Age at Sale column was added to the housing table. Its values were displayed alongside the existing property data, and the column's data type was set to a whole number.



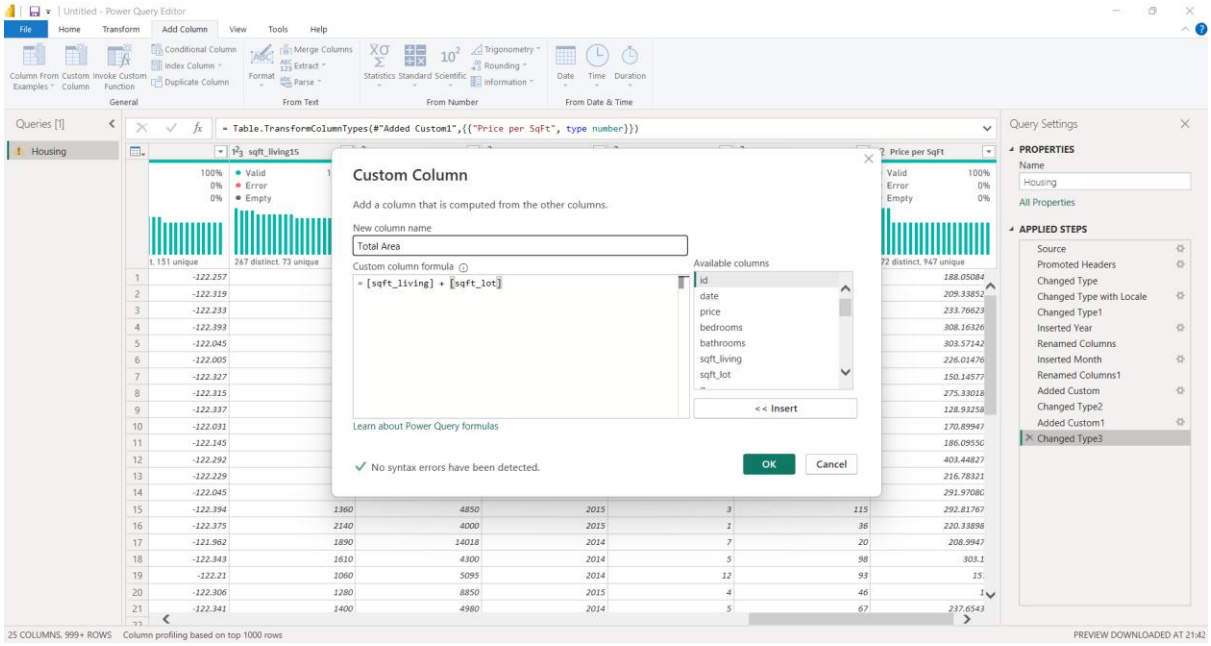
A custom column named Price per SqFt was created using the formula:

[price] / [sqft_living]

This calculates the price per square foot of living area for each property, enabling comparisons between properties of different sizes.



The Price per SqFt column was displayed with calculated values. Its data type was changed to a numeric type so that the resulting values could be used for calculations and analysis.

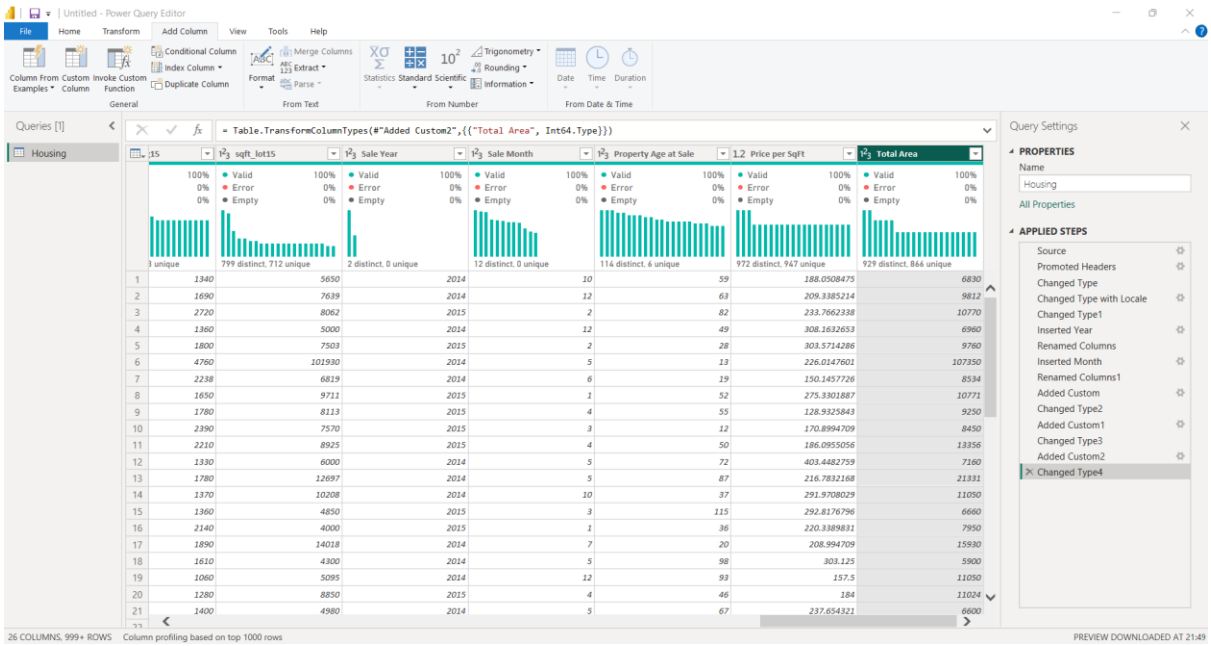


A custom column named Total Area was created using the formula:

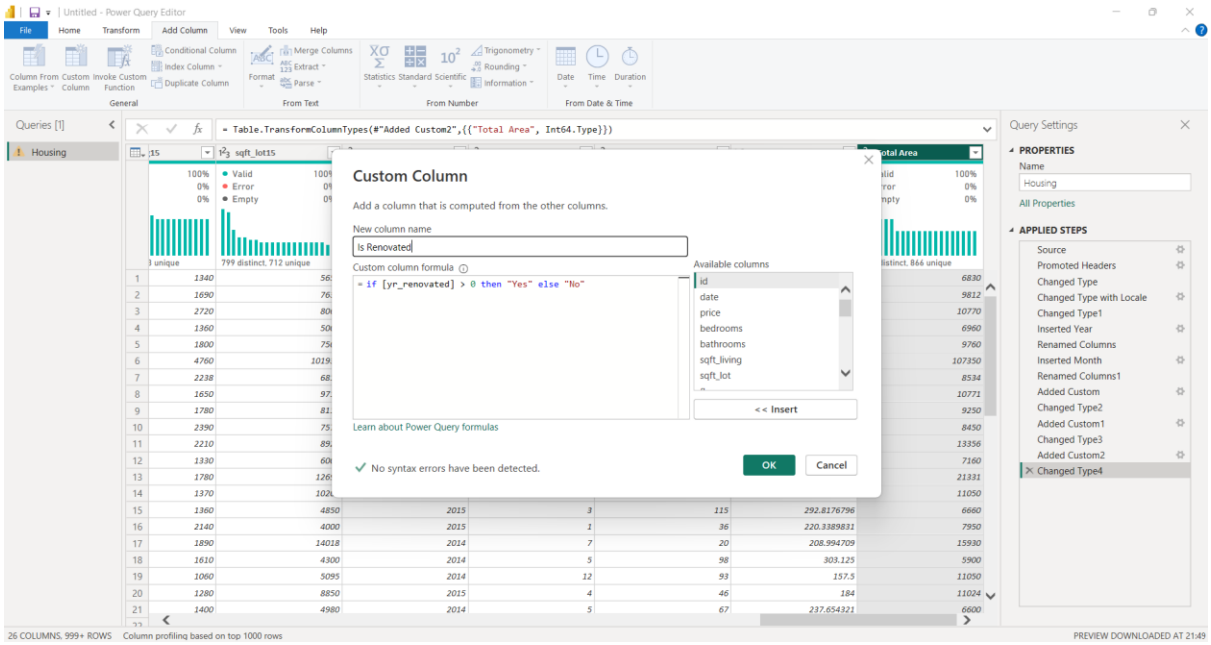
[sqft_living] + [sqft_lot]

This combines the living area and lot area into a derived metric for further analysis.

Note: This column represents the sum of living area and lot area, not the property's interior floor area alone.



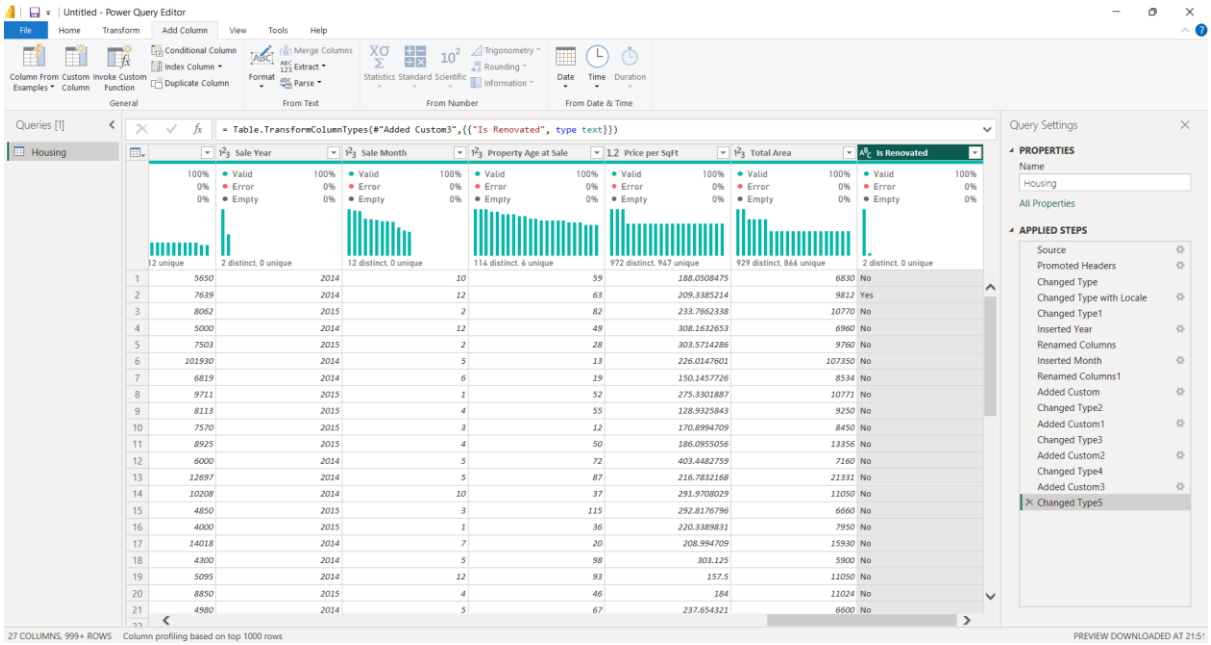
The Total Area column was added to the housing table, and the calculated values were displayed. The column's data type was set to a whole number for consistent numerical analysis.



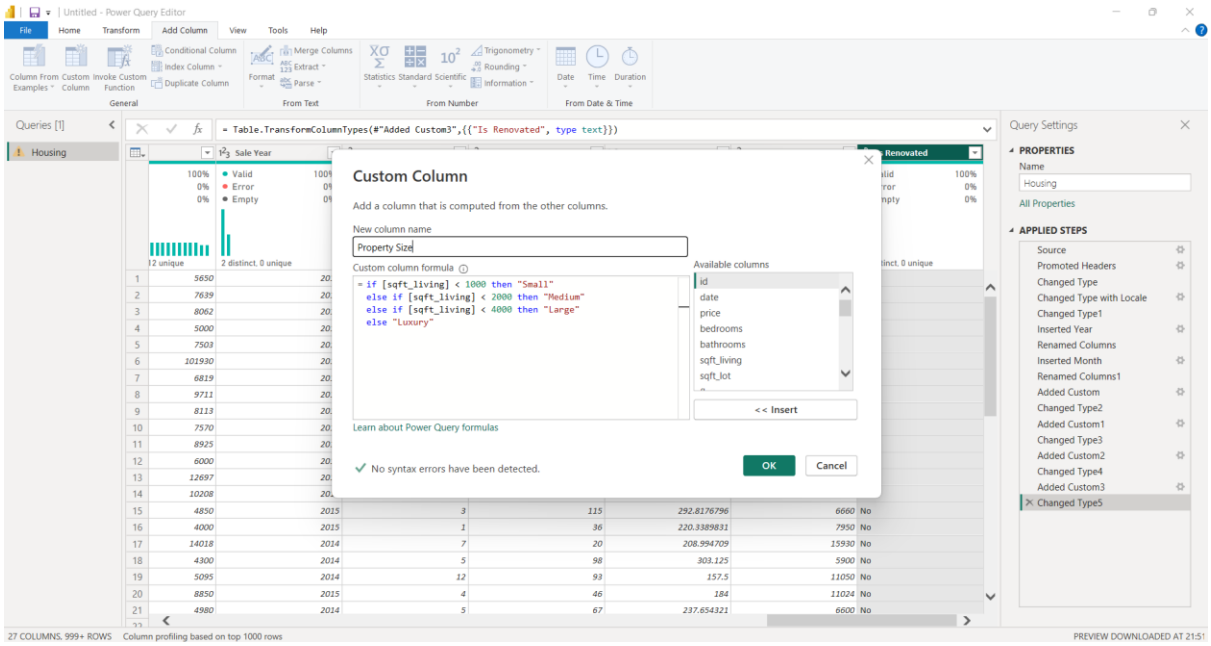
A custom column named Is Renovated was created using the formula:

if [yr_renovated] > 0 then "Yes" else "No"

This categorizes properties as renovated when the renovation year is greater than zero and as not renovated otherwise.



The newly created Is Renovated column was displayed with the values Yes and No. The column was assigned a text data type, making renovation status available for categorical analysis.



A custom column named Property Size was created to classify properties according to their living area.

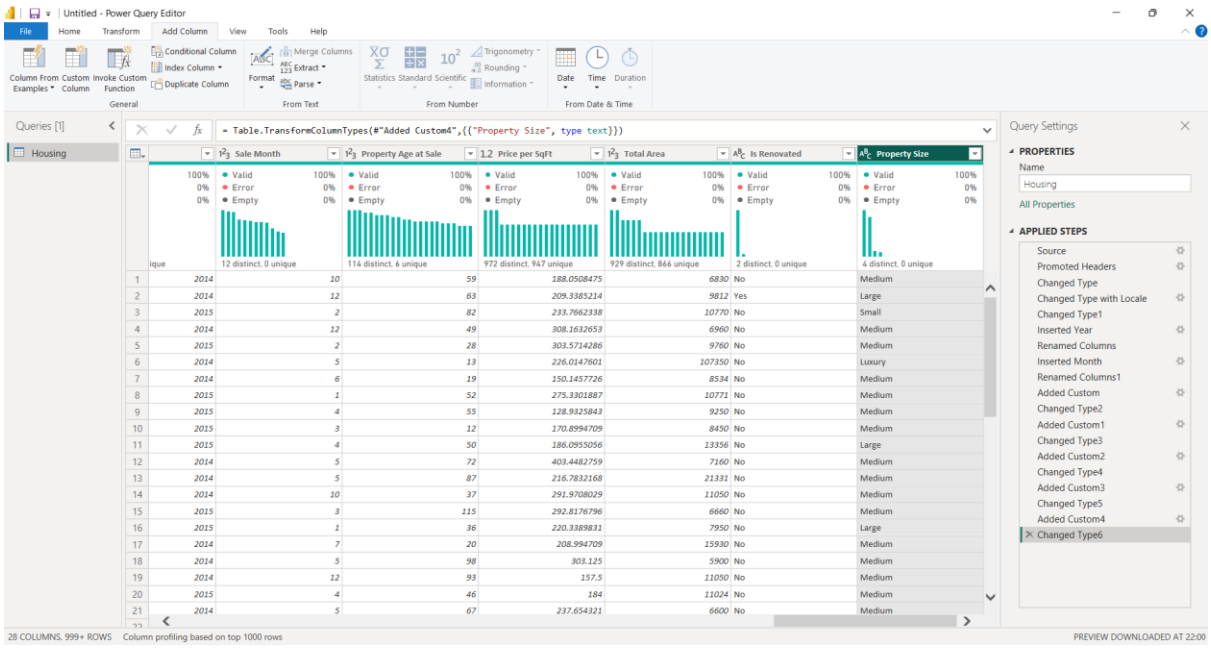
if [sqft_living] < 1000 then "Small"

else if [sqft_living] < 2000 then "Medium"

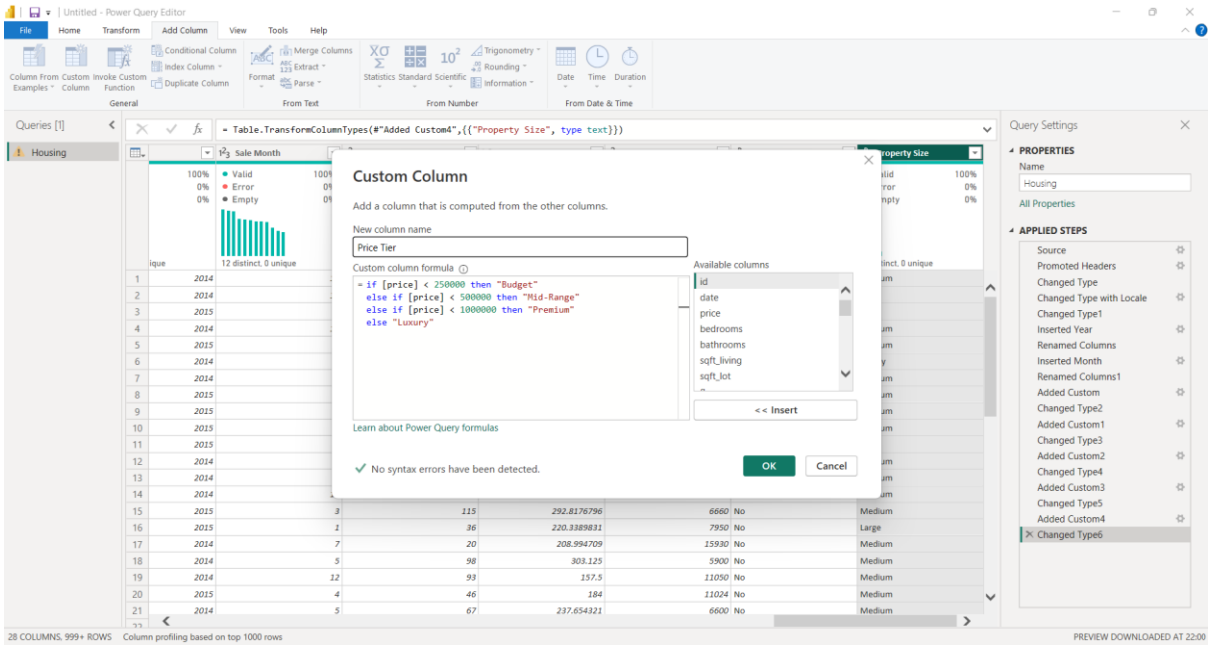
else if [sqft_living] < 4000 then "Large"

else "Luxury"

The classification uses the living-area thresholds shown in the formula to assign each property to a size category.



The Property Size column was added to the dataset. The resulting categories—Small, Medium, Large, and Luxury—were displayed alongside the other housing attributes.



A custom column named Price Tier was created to categorize properties based on their sale price.

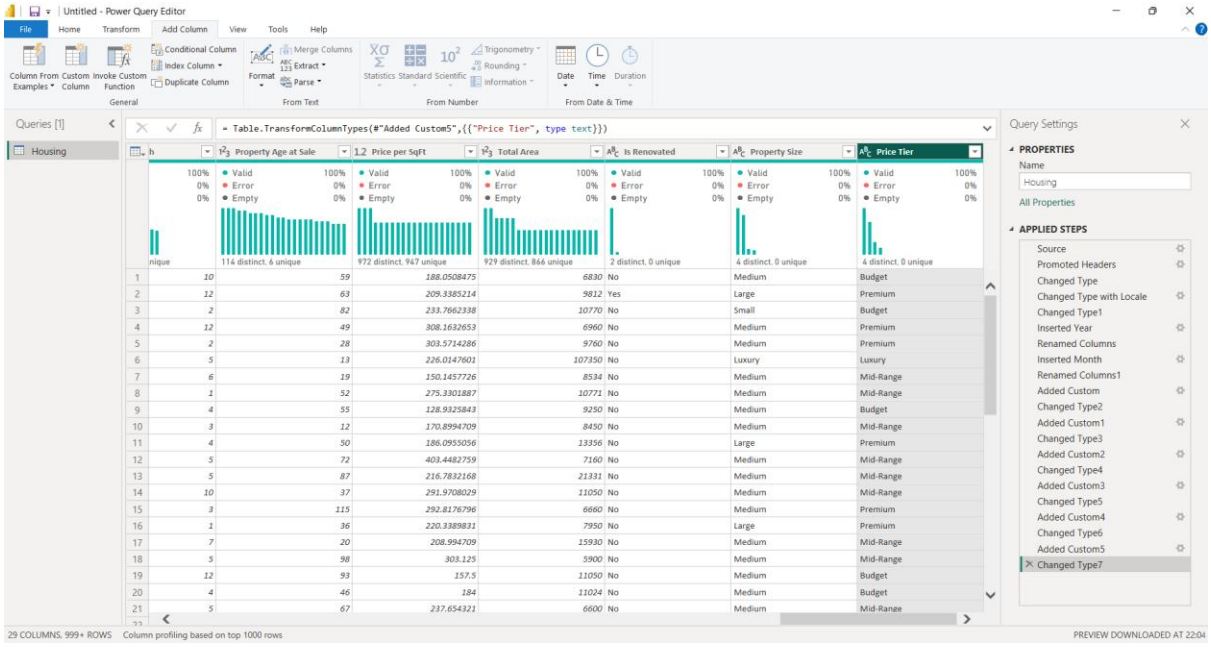
if [price] < 250000 then "Budget"

else if [price] < 500000 then "Mid-Range"

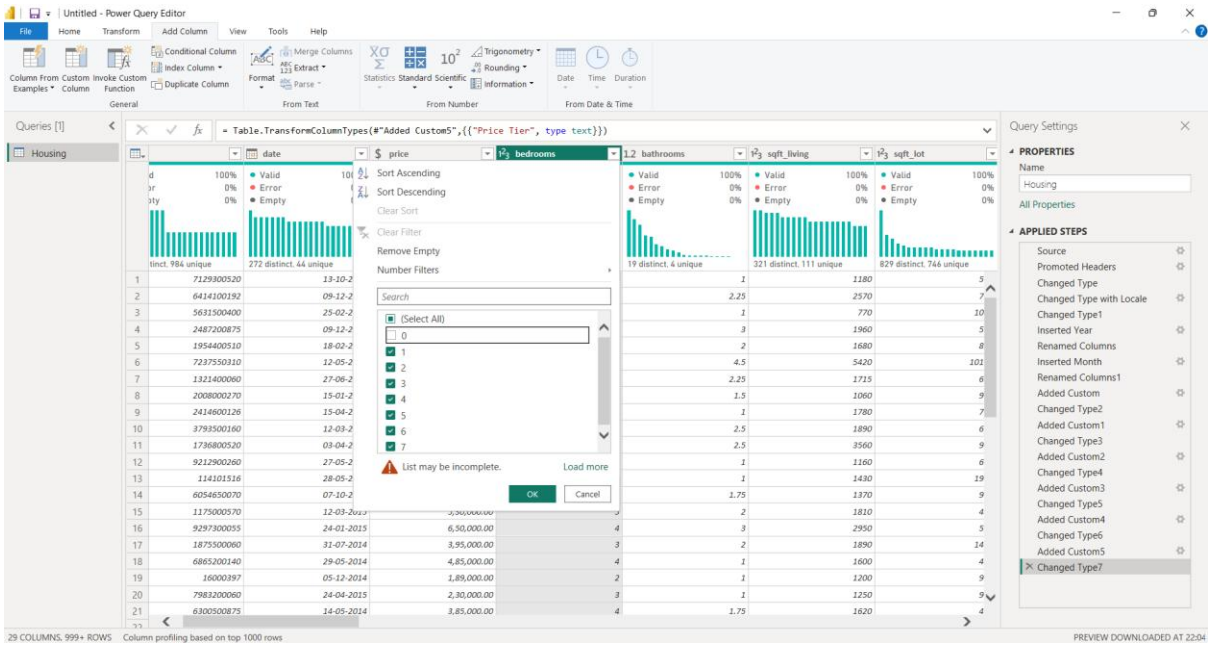
else if [price] < 1000000 then "Premium"

else "Luxury"

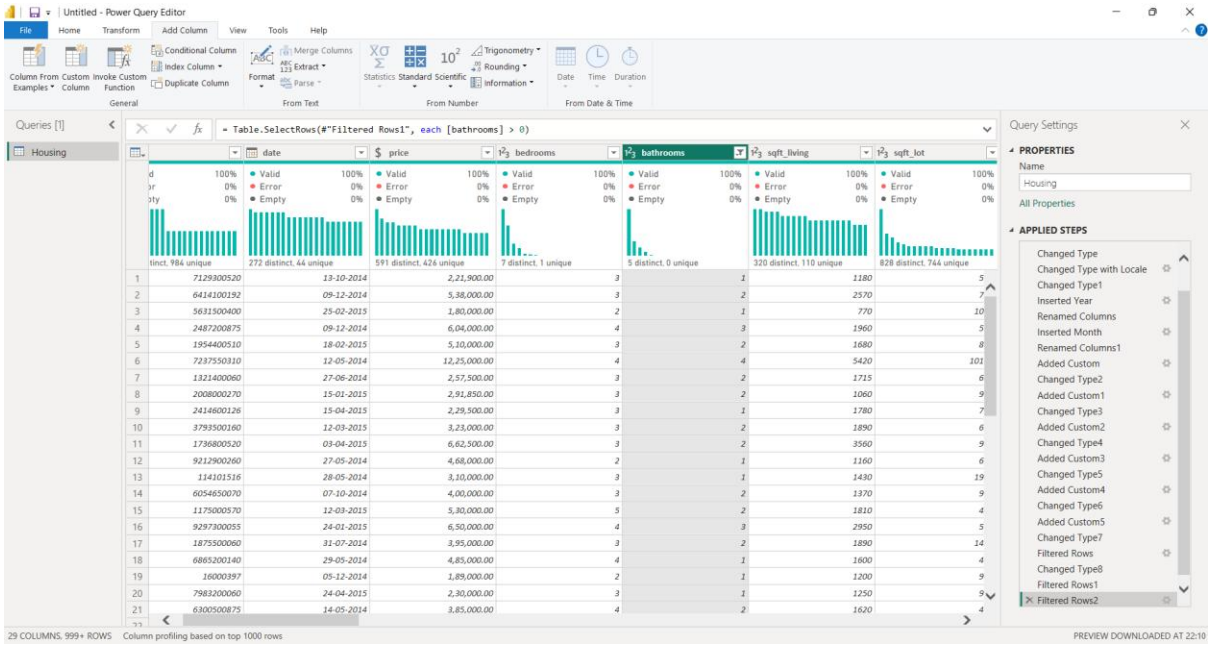
This groups properties into price categories using the specified thresholds.



The Price Tier column was added to the housing table, and the calculated categories were displayed. The column was set to a text data type for use in categorical analysis.



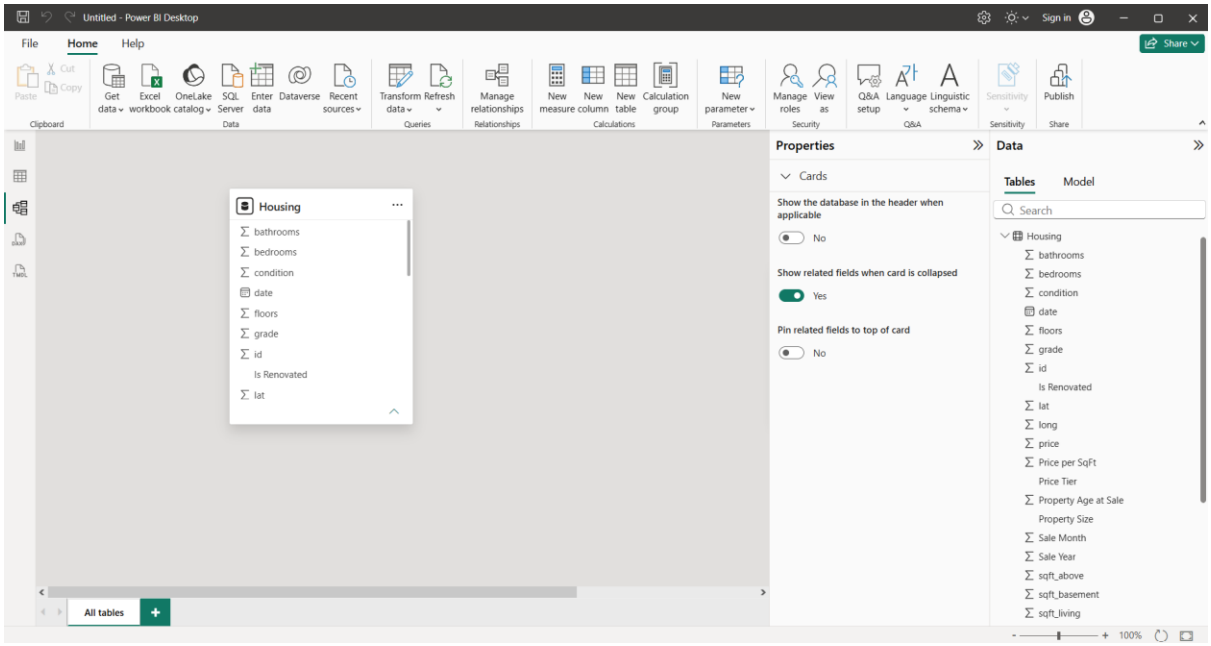
The filter menu for the bedrooms column was opened, and the value 0 was unchecked. This removes properties with zero bedrooms from the dataset, as required by the assignment.



A filter was applied to the bathrooms column using the condition:

[bathrooms] > 0

The screenshot shows the filter step in the Applied Steps pane, ensuring that properties with zero bathrooms are excluded.



The Power BI Model view was opened to review the Housing table and its available columns. The transformed fields, including Is Renovated, Price Tier, Property Age at Sale, Property Size, Sale Month, and Sale Year, were visible in the table.